HACKSMITHS

VEIL

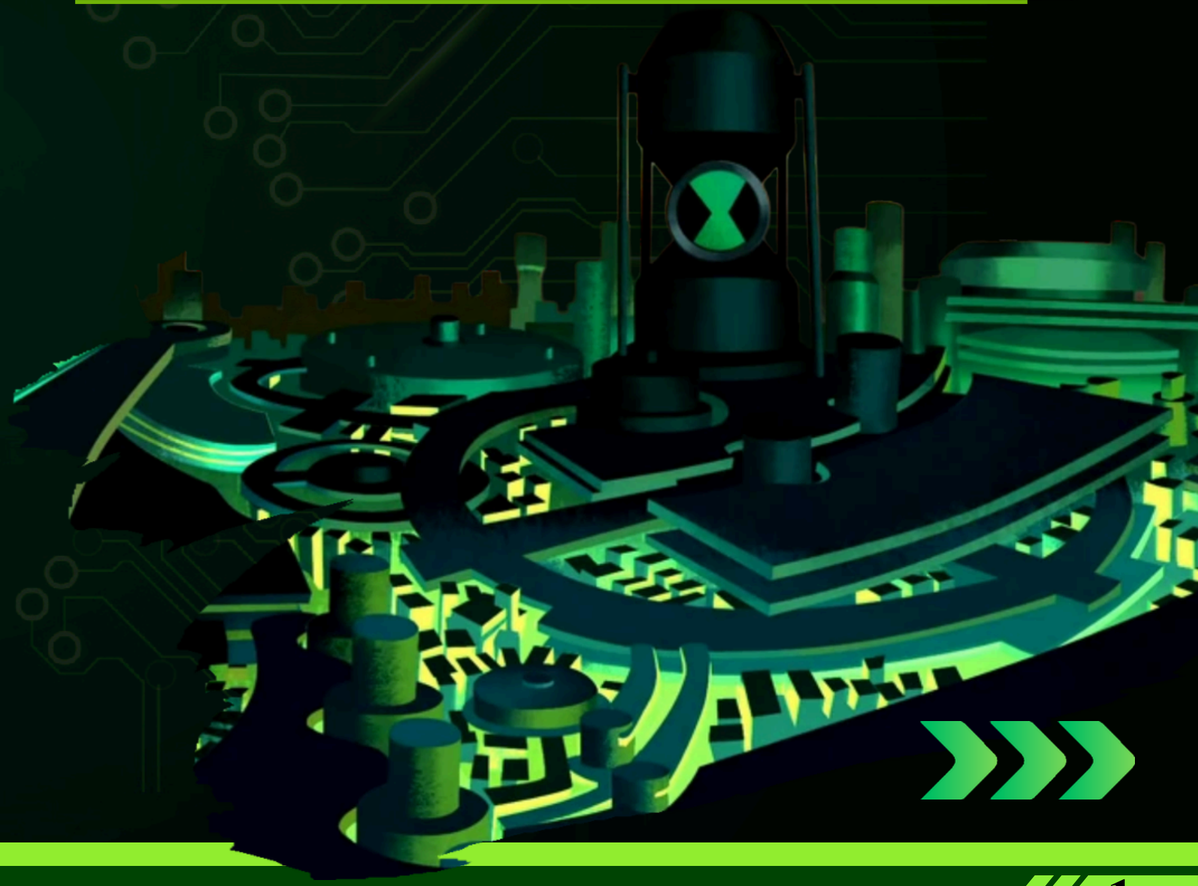
HACKBiOS
WHERE IDEAS BOOT INTO REALITY

Problem Statement

Users increasingly submit confidential information to AI systems and online services for processing, but lose visibility and control over how that information is handled beyond their immediate interaction.

Theme

Web3/BlockChain



Team Members

01 **Deepti Patel**

02 **Priti Sonkar**

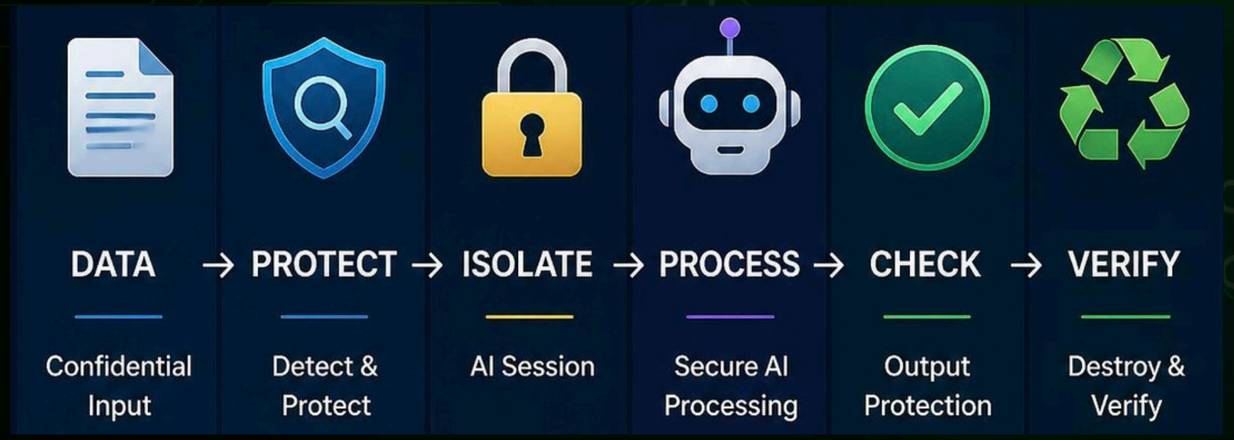
03 **Mohd Aatif Khan**

04 **Tushar**

VEIL

PROPOSED SOLUTION

- VEIL is a privacy-preserving AI platform that creates a temporary, policy-controlled environment for processing confidential information.
- Instead of sending sensitive data directly to an AI service and relying entirely on the provider's retention policies, VEIL Guard places a security layer between the user and the AI processing environment.



INNOVATION

Ephemeral AI Sessions

Processing occurs inside an isolated, short-lived environment.

Policy Before Processing

Sensitive information is detected and handled according to configurable rules.

Local Confidential AI

Ollama enables local inference without requiring confidential data to be sent to a third-party AI API.

Output Protection

VEIL also checks generated output for sensitive-data leakage.

Your paragraph text

TECHNICAL APPROACH

Confidentiality Assurance

Technology

FRONTEND
- Tauri • React • JavaScript

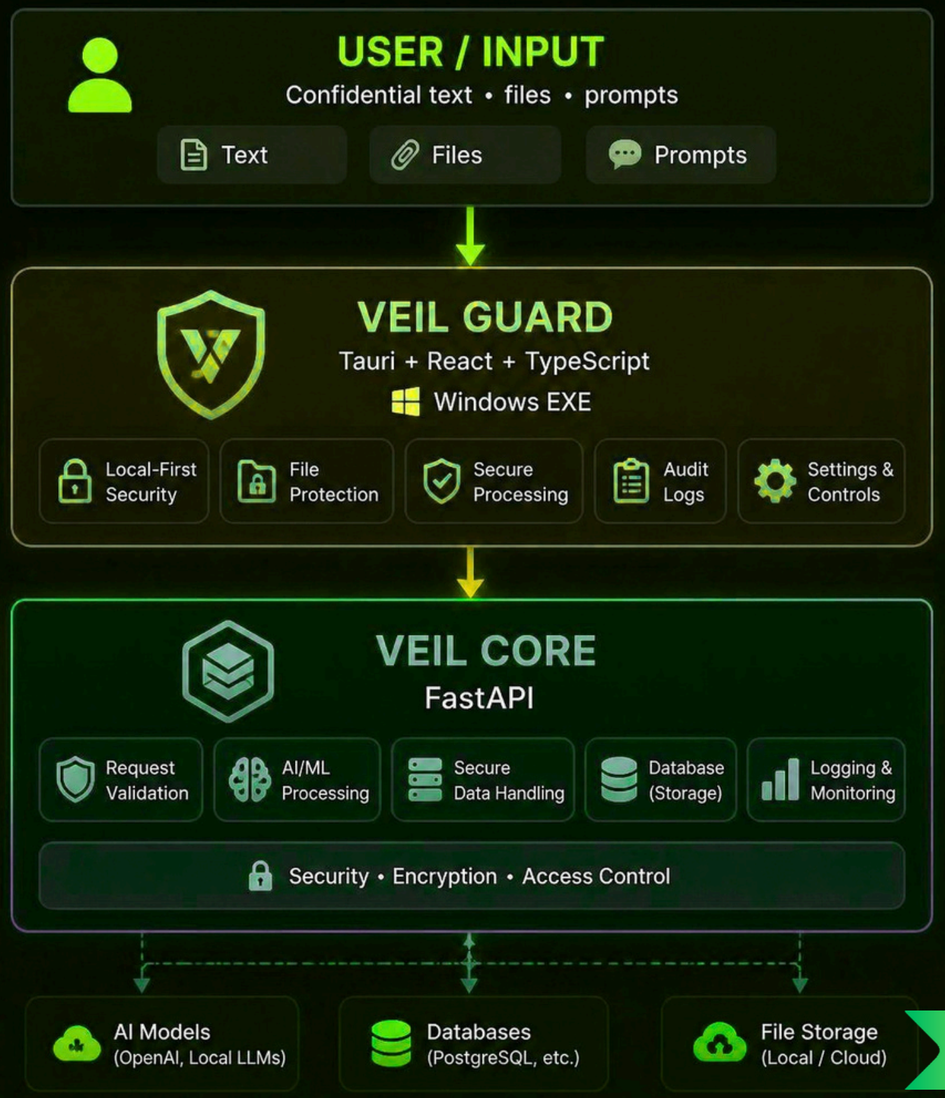
BACKEND
- FastAPI • Python • SQLAlchemy • PostgreSQL

SECURITY
- Presidio • Web Crypto • DLP • Policy Engine

ISOLATION & AI
- Docker • Ollama • Local LLM

WEB3
- Solidity • Polygon Testnet

No confidential plaintext, passwords, API keys or session keys stored in PostgreSQL.



HACKSMITHS

FEASIBILITY AND VIABILITY

HACKBIOS
WHERE IDEAS BOOT INTO REALITY

Data Leakage

- **Risk:** Confidential information may be exposed during AI processing.
- **Mitigation:** Presidio-based detection, policy enforcement and input/output DLP.

AI & Third-Party Exposure

- **Risk:** External AI services create additional data-retention and trust risks.
- **Mitigation:** Local Ollama-based processing keeps confidential workloads on user-controlled infrastructure.

Persistent Data

- **Risk:** Sensitive data or temporary artifacts may remain after processing.
- **Mitigation:** Ephemeral Docker containers, temporary storage and session-specific key destruction.

Blockchain Privacy

- **Risk:** Storing sensitive information on-chain creates permanent exposure.
- **Mitigation:** Polygon records only hashes and non-sensitive session/lifecycle metadata.

Viability

Technically practical, economically viable, and scalable.

OPEN-SOURCE & COST EFFECTIVE

Uses an open-source technology stack and Polygon testnet, keeping development and prototype costs low.

FOCUSED DEPLOYMENT

A Windows-first approach keeps the initial implementation focused, achievable and suitable for hackathon demonstration.

MODULAR & SCALABLE ARCHITECTURE

VEIL's security, AI and blockchain components are modular, allowing each layer to evolve independently.

LOCAL-FIRST & FUTURE READY

Local AI processing reduces dependency on external providers, while the shared VEIL Core enables future expansion to VEIL Connect browser extension.

HACKSMITHS

IMPACT AND BENEFITS

HACKBIOS
WHERE IDEAS BOOT INTO REALITY

Privacy & Data Control

- Minimizes persistent copies of confidential information
- Gives users greater control over how sensitive data is processed
- Session-based processing limits the data lifecycle

Verifiable Trust

- Blockchain provides tamper-evident lifecycle records
- Policies and session events can be independently verified
- Sensitive plaintext is never stored on-chain

Secure AI Adoption

- Enables AI processing for sensitive workloads
- Local-first inference reduces dependency on external AI providers
- Input and output DLP helps prevent accidental data leakage

Broad Real-World Applications

- **Banking • Healthcare • Legal**
- **Government • Enterprise • Research**

Confidential documents and information can be processed within a controlled environment without making permanent storage the default.

RESEARCH AND REFERENCES

Privacy & Data Protection

- **Microsoft Presidio —** PII detection, anonymization and data protection - ***microsoft.github.io/presidio***
- **NIST Privacy Framework —** Privacy-risk management principles - ***nist.gov/privacy-framework***

AI Security

- **OWASP Top 10 for LLM Applications —** LLM security threats and mitigations **- *owasp.org/www-project-top-10-for-large-language-model-applications***
- **Ollama Documentation —** Local LLM deployment and inference **- *docs.ollama.com***

Application Security & Isolation

- **Tauri Documentation —** Cross-platform desktop application architecture **- *tauri.app***
- **Docker Documentation —** Containerization and isolated execution - ***docs.docker.com***
- **Web Cryptography API — W3C —** Browser cryptographic primitives **- *w3.org/TR/WebCryptoAPI***

Blockchain & Verification

- **Polygon Documentation —** Blockchain infrastructure and smart contracts **- *docs.polygon.technology***

